

Advanced 3D Game Development

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CA3 - Networked 3D Game

Networking Models Overview

Client - Server

- ▶ Central authority
- ▶ Stable + secure
- ▶ Higher server cost

Peer to peer

- ▶ Low infrastructure cost
- ▶ Host advantage
- ▶ Security issues

Hybrid

- ▶ Shared simulation
- ▶ Flexible authority
- ▶ Common in modern middleware

My Fusion Implementation

Fusion Shared Mode

- InputAuthority per player
- Distributed StateAuthority
- Session-based replication

Implemented Features

- Session join
- Network spawning
- Player ownership
- Authentication flow



Authority Models (Fusion vs unreal)

Fusion

- ▶ InputAuthority
- ▶ StateAuthority
- ▶ Shared Mode

Unreal Engine 5

- ▶ Server
- ▶ Owning client
- ▶ Simulated proxy

Replication and Synchronisation

Fusion

- [Networked]
- Tick-based replication
- RPC support

Unreal

- Replicated UPROPERTY
- Server replication
- Multicast RPCs

```
public class PlayerStats : NetworkBehaviour
{
    [Networked]
     1 usage
    public int Health { get; set; }

    [Networked]
     1 usage
    public NetworkString<_16> PlayerName { get; set; }

     hillyleopard133
    public override void Spawned()
    {
        if (Object.HasStateAuthority)
        {
            Health = 100;
            PlayerName  = $"Player_{Runner.LocalPlayer.PlayerId}";
        }
    }
}
```

Lag and Latency

Sources of lag

- ▶ RTT
- ▶ Packet loss
- ▶ Jitter
- ▶ Tick delay

Problems

- ▶ Rubberbanding
- ▶ Input delay
- ▶ Desync

Compensation Techniques

Client Prediction

Local immediate movement

Interpolation

Smoothing remote movement

Server Reconciliation

Correcting incorrect prediction

Extrapolation

Guessing future state

Authentication + Access Control

Why authentication matters

- ▶ Prevent anonymous abuse
- ▶ Identify users
- ▶ Secure multiplayer services

My implementation

- ▶ Unity Authentication
- ▶ Anonymous sign-in
- ▶ Fusion integration

Industrial Expectations and Trends

Industry Expectations

- ▶ Authority understanding
- ▶ Replication systems
- ▶ Debugging skills
- ▶ Version control discipline

Emerging Trends

- ▶ Cloud-hosted multiplayer
- ▶ Large-scale sessions
- ▶ Hybrid authority systems
- ▶ Deterministic networking

Reflection

What went wrong

- ▶ Incorrect shared mode assumptions
- ▶ Callback ordering issues
- ▶ Session instability

What I'd change

- ▶ Better modularity
- ▶ Earlier networked state design
- ▶ More structured bebugging workflow